


# YEAR 10 SCIENCE

**AUSTRALIAN CURRICULUM V 7.5 ARC TRANSITION PROGRAM**
**Physical sciences**



The physical sciences sub-strand is concerned with understanding the nature of forces and motion, and matter and energy. The two key concepts developed within this sub-strand are that: forces affect the behaviour of objects; and that energy can be transferred and transformed from one form to another. Through this sub-strand students gain an understanding of how an object's motion (direction, speed and acceleration) is influenced by a range of contact and non-contact forces such as friction, magnetism, gravity and electrostatic forces. They develop an understanding of the concept of energy and how energy transfer is associated with phenomena involving motion, heat, sound, light and electricity. They appreciate that concepts of force, motion, matter and energy apply to systems ranging in scale from atoms to the universe itself.

In Year 10, students examine the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects.

## Content – Chemical Science

### Physical sciences

Energy conservation in a system can be explained by describing energy transfers and transformations (ACSSU190)

- recognising that the Law of Conservation of Energy explains that total energy is maintained in energy transfer and transformation
- recognising that in energy transfer and transformation, a variety of processes can occur, so that the usable energy is reduced and the system is not 100% efficient
- comparing energy changes in interactions such as car crashes, pendulums, lifting and dropping
- using models to describe how energy is transferred and transformed within systems

### The motion of objects can be described and predicted using the laws of physics (ACSSU229)

- gathering data to analyse everyday motions produced by forces, such as measurements of distance and time, speed, force, mass and acceleration

- recognising that a stationary object, or a moving object with constant motion, has balanced forces acting on it
- using Newton's Second Law to predict how a force affects the movement of an object
- recognising and applying Newton's Third Law to describe the effect of interactions between two objects

## Assessment Outcome

By the end of Year 10, students explain the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects.

Students develop questions and hypotheses and independently design and improve appropriate methods of investigation, including fieldwork and laboratory experimentation. They explain how they have considered reliability, safety, fairness and ethical actions in their methods and identify where digital technologies can be used to enhance the quality of data. When analysing data, selecting evidence and developing and justifying conclusions, they identify alternative explanations for findings and explain any sources of uncertainty. Students evaluate the validity and reliability of claims made in secondary sources with reference to currently held scientific views, the quality of the methodology and the evidence cited. They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes.

## Unit Organisation and Assessment Schedule

| Week | Content | Reference | Assessment |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
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| 11 | | | |

## Assessment Outline

| Assessment Type | Assessment Name | Timing | Weighting |
|---|---|---|---|
| Investigation | | | |
| Mid Topic Test | | | |

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|---|---|---|---|
| Test | | | |
| Experiment | | | |
| Extended Response | | | |
| Semester One Exam | | | |